Improving Solar Panel Efficiency Though Perovskite-Silicon Tandem Solar Cells

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Introduction

Solar energy is playing a big role in the world's switch to renewable electricity. As more people need electricity and countries want to cut down on gas emissions, photovoltaic technology is a great way to turn sunlight into electricity without burning fossil fuels. This has motivated engineers to find ways to improve the efficiency of solar panels. A more efficient panel can produce more electricity from the same amount of sunlight and physical space, allowing the better use of space and sun. Improving solar panel efficiency is therefore an important engineering goal because it can increase electricity production without the use of fossil fuels.

Silicon is the primary material used to make solar panels because it performs well and has a well-established manufacturing industry. More than 90% of today' s photovoltaic systems use crystalline silicon (National Renewable Energy Laboratory [NREL], n.d.). However, conventional silicon solar cells are approaching some fundamental efficiency limitations. Silicon cannot convert every part of the solar spectrum into electricity efficiently because it has a specific band gap. Some photons do not have enough energy to be absorbed, while other photons have more energy than the silicon needs and lose their excess energy as heat. Additional losses occur though reflection, electrical resistance, recombination, and other material limitations (Andreani et al., 2019; Kirchartz et al., 2025). These losses make it increasingly difficult to improve the efficiency of the conventional single junction-junction silicon cells.

One way to face these challenges is to create tandem solar cells that combine perovskite and silicon. Instead of relying on one material to absorb the entire solar spectrum, tandem solar cells use two semiconductor materials that can absorb different portions of sunlight. By placing a perovskite layer above a silicon cell, higher energy photons can be absorbed by the perovskite while lower-energy photons pass through to the silicon layer. Research has shown that tandem photovoltaic are making progress, with the U.S Department of Energy (DOE) reporting that perovskite-silicon tandem cells have achieved efficiencies of 34% (DOE, n.d-a)

Perovskite-silicon tandem cells are therefore a promising approach for improving solar-panel efficiency while still building on existing silicon technology. However, this technology is not ready for widespread commercial use yet. Other factors should be considered such as long-term

stability, moisture sensitivity, environmental effects, large scale manufacturing, cost, and outdoor performance under changing conditions. This paper examines how conventional silicon solar cells work, why their efficiency is limited, how perovskite-silicon tandem cells can overcome some of these limitations, and what engineering challenges must be addressed before tandem technology can become widely used.

Conventional Silicon Solar Cells

Sunlight is converted into electricity in a conventional silicon solar cell through the photovoltaic effect. When sunlight reaches the semiconductor, photons with enough energy can be absorbed by the silicon. This absorbed energy creates mobile charge carriers, including electrons and holes. Internal electric fields and selective electrical contacts help separate and collect these charge carriers, producing an electrical current. The current can then be transferred through an external circuit and used as electrical power.

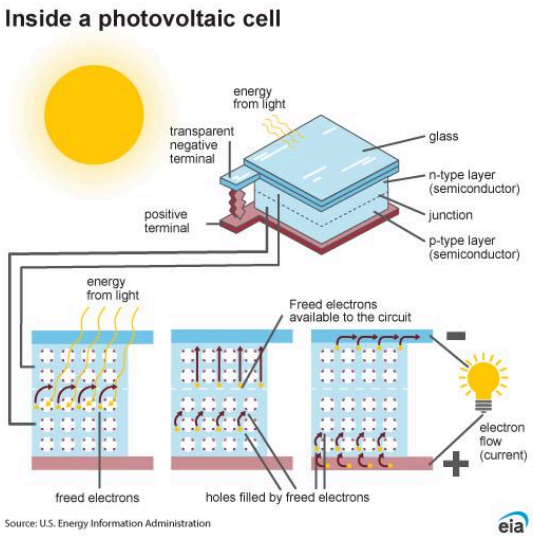


Figure 1. Basic structure and operation of a conventional silicon solar cell. Source: U.S. Department of Energy (DOE, n.d.-a)

Silicon is a popular material for solar cells because its properties are well understood and its manufacturing technology is highly developed. Andreani et al. (2019) explain that silicon is suitable for photovoltaic applications because of its band gap, availability, relatively low toxicity,

and mature technology. Decades of research have improved silicon-cell design, surface treatment, electrical contacts, and methods for managing light. As a result, crystalline silicon remains the foundation of most photovoltaic systems.

Although silicon is very popular, its maturity also creates a challenge for engineers. Improving an established technology becomes increasingly difficult as it approaches its theoretical limits. A single-junction silicon solar cell has a fundamental efficiency limitation known as the Shockley-Queisser limit. This limit describes the maximum theoretical efficiency of a single-junction semiconductor when it is used to turn sunlight into electricity under idealized conditions. Andreani et al. (2019) discuss this limit and explain how optical and material strategies can help real silicon cells approach higher efficiencies, but they cannot completely eliminate the fundamental limitations of a single-junction cell.

Factors Limiting Silicon Solar-Cell Efficiency

Several different losses contribute to the efficiency limitations of conventional silicon solar cells. One of the most important is the mismatch between the energy distribution of sunlight and the single band gap of silicon. Low-energy photons may not have enough energy to generate useful charge carriers, while higher-energy photons lose part of their energy through thermalization. Kirchartz et al. (2025) describe incomplete sunlight absorption as well as optical, electrical, and material losses as important factors limiting photovoltaic performance. These different losses demonstrate why improving only one component of a silicon cell cannot completely solve the overall efficiency problem.

Thermalization is one of the major sources of energy loss. The problem is that sunlight contains photons with many different energies, while silicon has one primary band gap. For a photon to contribute to electricity generation, it must have enough energy to excite an electron across the band gap. If a photon has less energy than the bandgap, it cannot be efficiently absorbed and may pass through the material without contributing useful electrical energy. On the other hand, when a photon has significantly more energy than the band gap, its excess energy is not converted into additional electrical energy. Instead, that excess energy is primarily lost as heat. Therefore, even a highly optimized silicon solar cell cannot efficiently use every part of the solar spectrum.

Reflection is another important problem. Not all sunlight that reaches a solar panel enters the semiconductor. Some light is reflected from the surface before it can be absorbed. Researchers have therefore developed different methods of surface texturing and light trapping to reduce reflection and increase the amount of light absorbed by silicon. Saive (2021) reviews several light-trapping techniques, including surface pyramids, photonic structures, nanostructures, and external reflective structures. With these techniques, the optical path length can give light more opportunities to be absorbed.

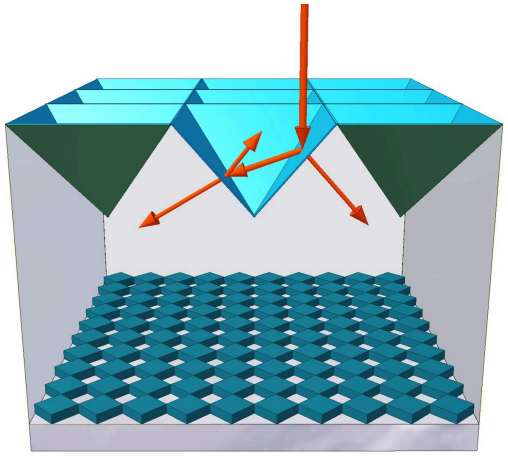


Figure 2. Pyramid surface texturing and Lambertian light trapping reduce reflection losses by increasing the optical path length inside silicon. Based on Saive (2021) and Andreani et al. (2019).

Lambertian light trapping is an important concept in improving the absorption of sunlight. A Lambertian surface scatters in many directions instead of allowing it to travel directly out of the cell. This increases the distance the light travels though the silicon and therefore increases the probability that it will be absorbed. . Andreani et al. (2019) discuss Lambertian scattering as a theoretical benchmark for light trapping and explain how it can increase the optical path length within silicon. Although advanced light trapping can reduce the optical losses, it can solve the fundamental spectral mismatch of a single-junction silicon cell.

Electrical and material losses also reduce photovoltaic efficiency. Even when sunlight is absorbed and charge carriers are created, not all of those carriers contribute to useful electrical output. Electrons and holes can recombine before reaching the electrical contacts, which reduces the amount of current that can be collected. Electrical resistance and imperfect interfaces can also cause additional losses. Kirchartz et al. (2025) emphasize that improving

photovoltaic efficiency requires addressing multiple optical, electrical, and material losses rather than relying on a single technological improvement.

These limitations explain why researchers are exploring ways to combine silicon with another semiconductor material. Rather than forcing silicon to efficiently absorb every part of the solar spectrum, a second material can be added to absorb wavelengths that silicon cannot use as effectively.

Perovskite Material and Tandem Solar Cells

Perovskites have become an important area of solar-energy research because their material properties can be adjusted for photovoltaic use. Perovskites have a particular crystal structure, and their chemical composition can be modified to change their optical and electrical properties. This tunability allows researchers to design perovskite materials with band gaps that complement silicon (DOE, n.d. -a).

One of the most important advantages of perovskites is their ability to absorb higher-energy portions of sunlight while allowing lower-energy photons to pass through. This makes them particularly useful as the top layer of a tandem solar cell. The perovskite can absorb photons that would otherwise lose a large amount of their energy through thermalization in silicon. Meanwhile, lower-energy photons can travel through the perovskite and reach the silicon bottom cell. By dividing the solar spectrum between the two materials, the tandem structure can reduce some of the energy losses that occur when silicon is used alone.

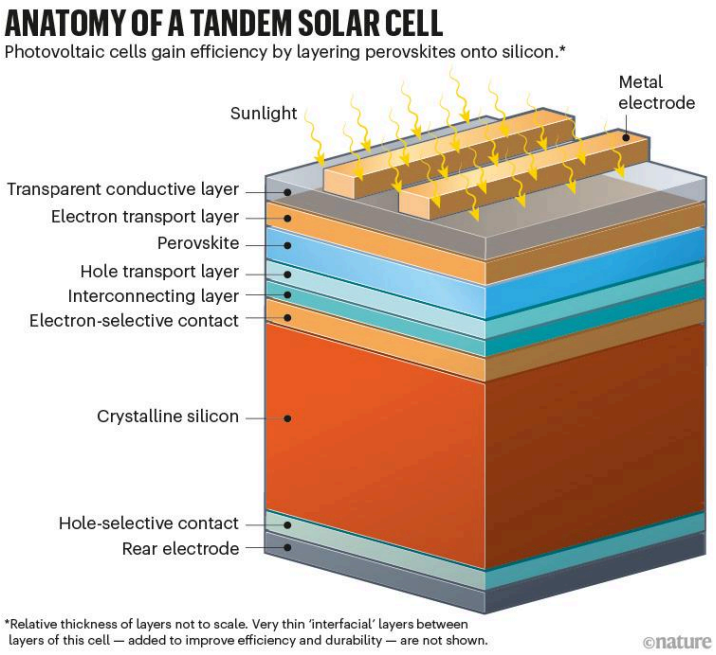


Figure 3. Layer structure of a perovskite-silicon tandem solar cell showing how different semiconductor layers absorb different portions of sunlight. Source: U.S. Department of Energy (DOE, n.d.-a).

Tandem cells can be constructed using different electrical configurations. In a two-terminal tandem cell, the top and bottom cells are electrically connected so that they operate together as a single device. Other configurations use separate electrical connections. Two-terminal architectures are particularly interesting because they may be compatible with existing photovoltaic manufacturing and can simplify integration into solar systems. Researchers have therefore been studying device architecture, material stability, and methods for depositing perovskite layers onto large silicon substrates.

The rapid progress of this technology demonstrates its potential. The DOE reports that perovskite-silicon tandem cells have reached efficiencies approaching 34% in small-area research devices (DOE, n.d.-a). NREL's Best Research-Cell Efficiency Chart also tracks perovskite-silicon tandem technology as a major photovoltaic research category.

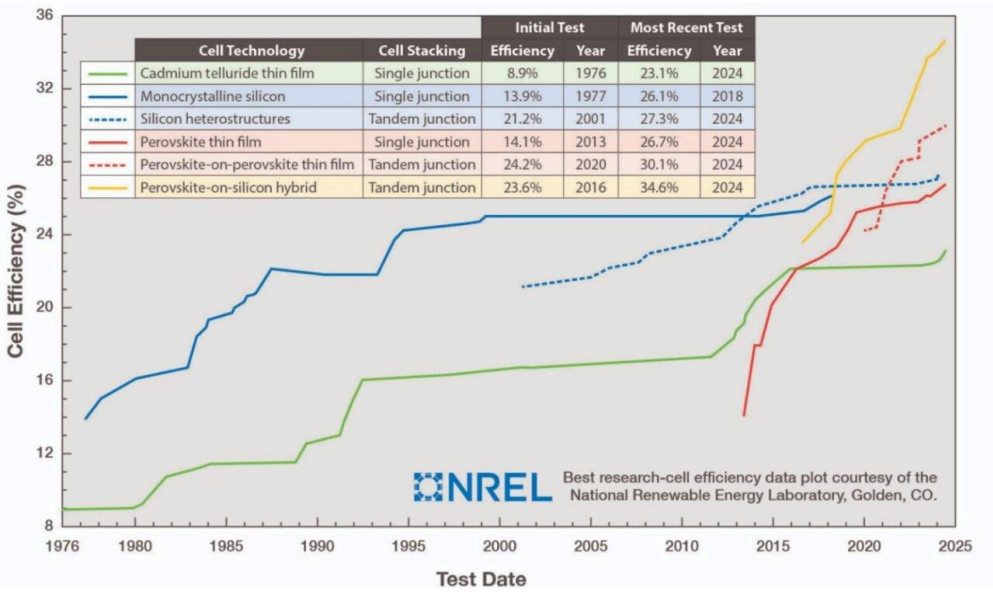


Figure 4. *Best research-cell efficiencies for photovoltaic technologies.*Source: National Renewable Energy Laboratory (NREL, n.d.).

Advantages and Performance of Tandem Solar Cells

The main advantage of perovskite-silicon tandem cells is their potential for higher conversion efficiency. A more efficient solar panel can produce more electricity from the same amount of sunlight and approximately the same physical area. Higher efficiency can also reduce the number of panels, mounting structures, electrical connections, and other components required to produce a certain amount of electricity.

Research has demonstrated the performance advantage of tandem cells under both laboratory and outdoor conditions. Chojniak et al. (2024) conducted outdoor measurements of a full-size bifacial perovskite-silicon tandem module under different spectral and environmental conditions. Their results demonstrated that tandem technology can achieve strong performance under realistic outdoor conditions, although changes in spectral distribution can affect efficiency.

Real-world testing is important because solar panels do not operate under perfectly controlled laboratory conditions. The spectrum of sunlight changes with weather, time of day, temperature, humanity, and geographic location.

There are also cost advantages in tandem cells. Messmer et al. (2021) compared different silicon bottom-cell technologies for perovskite-silicon tandems and found that some silicon

architectures can provide strong combinations of efficiency and cost. This demonstrates that improving the bottom silicon cell is just as important as improving the perovskite top cell. Another advantage is that perovskite-silicon technology could build on the existing silicon photovoltaic tech rather than requiring a completely new manufacturing system. The DOE identifies tandem photovoltaic as a promising approach because they can combine high efficiency with low cost of manufacturing.

Engineering Challenges

Although perovskite-silicon tandem cells have high efficiency, they still face several major engineering challenges. One of them most important is long-term stability. Commercial solar panels are expected to operate outdoors for many years, so a technology cannot be considered successful simply because it is efficient when new. It must maintain its performance after prolonged exposure to sunlight, heat, humidity, and other environmental conditions.

Perovskite materials can be sensitive to the above environmental conditions. The DOE identifies stability and durability, efficiency at larger scales, manufacturability, and validation as important challenges for perovskite photovoltaic solar panels (DOE, n.d.-a). Researchers are therefore investigating improved material compositions, protective coatings, encapsulation methods, and manufacturing techniques that can reduce degradation.

Large-scale manufacturing is another challenge. A small laboratory cell can be carefully produced under controlled conditions, but manufacturing large modules requires consistent deposition over a much larger area. The perovskite layer must be deposited uniformly while maintaining the appropriate optical and electrical properties. Small defects that may have little effect on a laboratory prototype solar panel can become much more significant when the solar panel is ready for commercial use.

Environmental impact is another important factor. Some perovskite solar cells contain lead, which creates concern about toxicity during manufacturing, operation, and disposal. Rodriguez-Garcia et al. (2024) conducted a life-cycle toxicity assessment of perovskite-silicon tandem photovoltaics and found that lead emissions and end-of-life management need to be considered.

Outdoor spectral conditions are another engineering challenge. The performance of a tandem cell depends on how sunlight is distributed between the perovskite and silicon absorbers.

Chojniak et al. (2024) found that changes in spectral distribution can influence tandem-cell efficiency. This means that laboratory efficiency alone does not provide a complete picture of how a tandem cell will perform in different environments. Engineers must evaluate energy production under realistic outdoor conditions.

Conclusion

Improving solar panel-efficiency is important because increasing the amount of electricity generated for each panel can make solar power more productive without requiring proportional increases in land, infrastructure, or panel area. Conventional crystalline silicon technology has become highly successful because it is efficient, well understood, eco-friendly, and supported by mature manufacturing processes. However, silicon is approaching the fundamental limitations of single-junction photovoltaic technology. Losses caused by the solar spectrum, thermalization, incomplete absorption, reflection, and electrical processes prevent a single silicon absorber from converting all incoming sunlight into electricity efficiently.

Perovskite-silicon tandem solar cells offer a promising solution becUse they combine two different semiconductor materials. The perovskite top layer can be adjusted to absorb higher-energy portions of the solar spectrum while allowing lower-energy photons to reach the silicon layer below. This division of the solar spectrum can reduce energy losses and produce substantially higher efficiencies than conventional single-junction silicon cells. Laboratory research, NREL efficiency data, and outdoor testing have all demonstrated the potential of this technology (Chojniak et al., 2024; DOE, n.d.-a; NREL, n.d.).

However, tandem cells are not ready yet to completely replace traditional silicon panels. Several engineering challenges remain, including long-term stability, moisture sensitivity, large-scale manufacturing, outdoor performance, cost, and environmental impact. The use of lead in many perovskite materials also means that responsible manufacturing, recycling, and disposal methods must be developed alongside efficiency improvements (Rodriguez-Garcia et al., 2024).

Perovskite-silicon tandem solar cells are particularly promising because they address a major weakness of traditional silicon cells: their inability to effectively use the solar solar spectrum. If engineers find ways to face the challenges of perovskite-silicon tandem solar cells, this breakthrough could significantly increase the amount of electricity. Although the technology still requires further development before widespread commercialization, combining the established

advantages of silicon with the spectral flexibility of perovskites makes tandem photovoltaics one of the most promising directions for the future of solar energy.

Word count: 2462

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